

Asher Siddiqui

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EDUCATION

FAST - NUCES *Bachelor of Software Engineering (6th semester)* 2021 - Present

Key Courses: Object Oriented Programming, Data Structures, Operating Systems, Design and Analysis of Algorithms, Database Design, Computer Networks, Web Engineering, Software Design and Architecture. Software Quality Assurance

SKILLS

Technical: Javascript, ExpressJS, NodeJS, Go Lang, REST APIs, Java, Spring Boot, SQL, Git, Bash, Linux, Docker, HTML

Soft: Organization, Time Management, Communication, Persistence, Problem Solving, Willingness to learn

WORK EXPERIENCE

FAST - NUCES

Jan 2024 - Present

Teaching Assistant - Operating Systems

- Collaborated efficiently with the instructor, assessing quizzes and assignments, showcasing strong communication and time management skills.
- Programmed an assignment evaluation tool, producing significant time savings and highlighting problem solving skills.

PROJECTS

github.com/pluuto19

Load Balancer [Go Lang, ExpressJS, Docker, Git, HTML]

- Designed and implemented a high-throughput HTTP reverse proxy in Go lang, integrating round-robin and least connection algorithms to efficiently distribute incoming traffic.
- Utilized goroutines and buffered channels, reducing computation overhead and enabling the application to be scalable, resulting in a throughput of 10,000 requests in 5 seconds.
- Implemented health checks to ensure the availability of backend servers, enhancing the reliability of the load balancer.

Project Management System [ExpressJS, MySQL, Handlebars, Git, HTML]

- Developed a project management system's webserver architecture using Express.js and SQL databases, adhering to a microservice architecture with three distinct services, each exposing its API.
- Orchestrated communication between services by integrating the controller with backend APIs, facilitating seamless interaction and data flow within the application.
- Employed API keys and JWT authentication with the REST APIs and controller, respectively, ensuring secure access across the system's services.

Visual Convex Hulls [Java, JavaFX, Git]

- Translated research papers into applications using JavaFX, demonstrating practical skills.
- Implemented and executed various convex hull algorithms, conducting an analysis of their computational complexity.

Miscellaneous [Python, Bash, ExpressJS, SQL, Java, Java Swing, Git, HTML]

- **Pyonic Reader:** A Python CLI tool for parsing PDFs, bolding initial letters of every word to help an ADHDer's reading focus.
- **Assess Buddy:** A bash tool for streamlined extraction, compilation, and execution of compressed folders, facilitating evaluation of student assignments.
- **Rent a Car & Autoshop:** An Express-based app with a normalized MySQL database for efficient CRUD operations.
- **A Night in the Woods:** A 2D platformer game made entirely in Java Swing, multithreaded to support background audio.

LEADERSHIP EXPERIENCE

Microsoft Learn Students Ambassador

Oct 2023 - Present

Technical Coordinator

- Coordinated technical activities for a workshop with approximately 50 participants. curating and managing workshop materials for the audience.
- Actively assisted the event management team during the workshop, ensuring smooth execution.

VOLUNTEER EXPERIENCE

Team Guide : Mentored K-12 students at Destination Imagination Pakistan 2024, fostering creativity and problem-solving skills, while facilitating campus navigation. Contributed to a positive team experience, resulting in strengthened leadership and mentorship abilities.

AWARDS & HONORS

Dean's List of Honors - Fall 2023